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BIOLOGY
STANDARD LEVEL
PAPER 1

SPECIMEN PAPER

45 minutes

INSTRUCTIONS TO CANDIDATES

- Do not open this examination paper until instructed to do so.
- Answer all the questions.
- For each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.
- The maximum mark for this examination paper is *[30 marks]*.

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1. Why is stem cell research considered unethical by some groups?

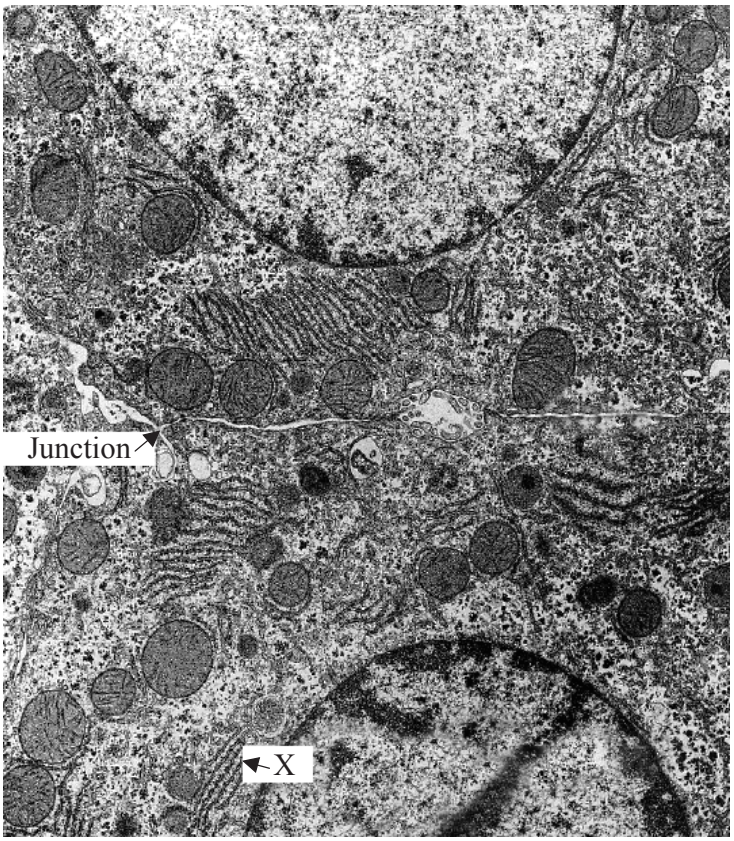
A. Stem cells are living organisms.

B. New organisms could be produced from stem cells.

C. Use of stem cells could involve the culture of pluripotent cells.

D. Use of embryonic cells involves the death of early-stage embryos.

2. The electron micrograph shows part of two adjacent liver cells.



[Source: adapted from www.relfe.com/Images/ratlivercells.gif]

What is happening at the structure labelled X?

A. Synthesis of proteins

B. Transport of proteins to the nucleus

C. Modification of proteins prior to export

D. Secretion of proteins through the plasma membrane

3. What is the approximate thickness of the plasma membrane of a cell?

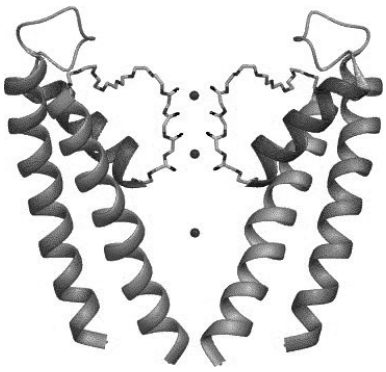
A. 10 μm

B. 50 μm

C. 10 nm

D. 50 nm

4. The diagram shows a potassium channel in an axon membrane. The three dots in the channel represent potassium ions.



[Source: adapted from S Y Noskov and B Roux, (2006), *Biophysical Chemistry*, **124** (3), pages 279–291]

What is the function of this potassium channel?

A. Facilitated diffusion of potassium ions into the axon

B. Passage of potassium ions by active transport out of the axon

C. Facilitated diffusion of potassium ions out of the axon

D. Passage of potassium ions by active transport into the axon

5. In the experiments performed by Meselson and Stahl, *E. coli* were grown for many generations in ^{15}N then for one generation in ^{14}N. What results for the DNA of the last generation showed that replication was semi-conservative?

A. Both strands containing only ^{15}N

B. Both strands containing only ^{14}N

C. One strand containing only ^{15}N and one stand containing only ^{14}N

D. Both strands containing a mixture of ^{15}N and ^{14}N in equal amounts

6. Cyclins were discovered by Timothy R Hunt in 1982 while studying sea urchins. What is a function of cyclins?

A. Circulation of seawater for gas exchange

B. Rotation of tentacles

C. Control of the cell cycle

D. Recycling of nutrients

7. In 1828, Friedrich Wöhler artificially produced urea (organic compound) by an internal rearrangement of the atoms of ammonium cyanate (inorganic compound). What important principle did this experiment show?

A. Organic compounds can be synthesized without the need of a vital force.

B. Organic compounds can only be formed from pre-existing inorganic compounds.

C. Inorganic compounds are essential for living organisms.

D. Organic compounds are more simple substances than inorganic compounds.

8. What promotes genetic variation in meiosis?

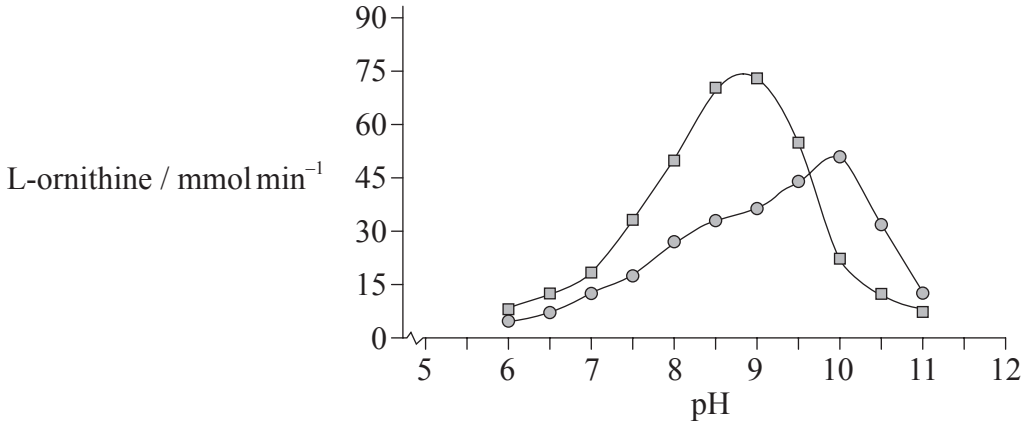
A. Cytokinesis

B. Replication of DNA

C. Condensation of chromosomes

D. Crossing over in homologous chromosomes

9. Arginase catalyses the hydrolysis of L-arginine to form L-ornithine and urea. Arginase was purified from cowpea cotyledons (●) and buffalo liver (■) and its activity was measured at different pH values.



[Source: adapted from S Dabir, *et al*., (2005), *International Journal of Biological Sciences*, **1** (3), pages 114–122]

What can be concluded from these results?

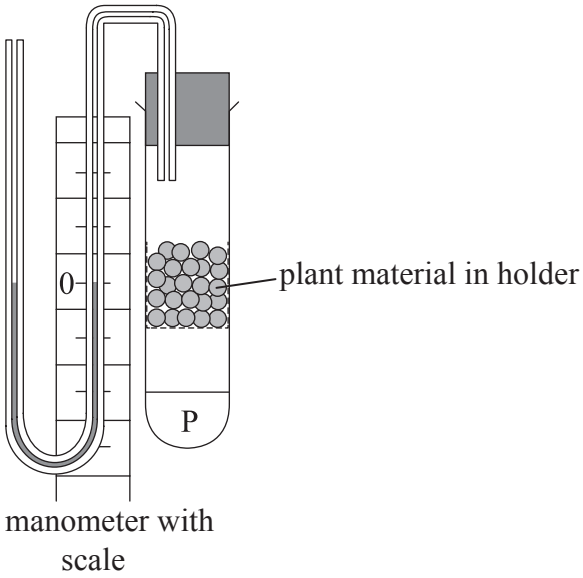
A. The optimum pH for buffalo liver arginase is 10.

B. Cowpea arginase works better in alkaline conditions.

C. Cowpea arginase works at a wider range of pH than buffalo liver arginase.

D. At a more acid pH the cowpea arginase works better than buffalo liver arginase.

10. The diagram shows seeds in a respirometer. What is substance P?



[Source: adapted from K Sands, (1998), *Problems in plant physiology*]

A. An acid used to absorb O_2 produced by seeds.

B. An acid used to release O_2 for use by seeds.

C. An alkali used to absorb CO_2 produced by seeds.

D. An alkali used to release CO_2 for use by seeds.

11. What is the energy absorbed by chlorophyll used directly for in plants?

I. To produce ATP

II. To split water

III. To fix CO_2

A. I only

B. III only

C. I and II only

D. II and III only

12. Which statement is part of the cell theory?

A. All cells have a cell wall.

B. Every cell shows emergent properties.

C. All cells come from pre-existing cells.

D. Every cell carries out all the functions of life.

13. The following alignment represents part of the sequence of a gene in two species, the mouse (*Mus musculus*) and woolly monkey (*Lagothrix lagotricha*).

Mouse MGDVEKGKKIFVMKCAQCHTVEKGGKHKTGPNLHGLFGRKTGQAAGFSYTDANKNK

Woolly monkey MGDVEKGKRIFIMKCSQCHTVEKGGKHKTGXNLHGLFGRKTGQASGYTYTEANKNK

What term is used for different forms of a gene such as these?

A. Loci

B. Alleles

C. Homologues

D. Heterologues

14. The rate of mutation in genes is fairly constant over time. What can be deduced from differences in amino acid sequences between particular proteins in different species?

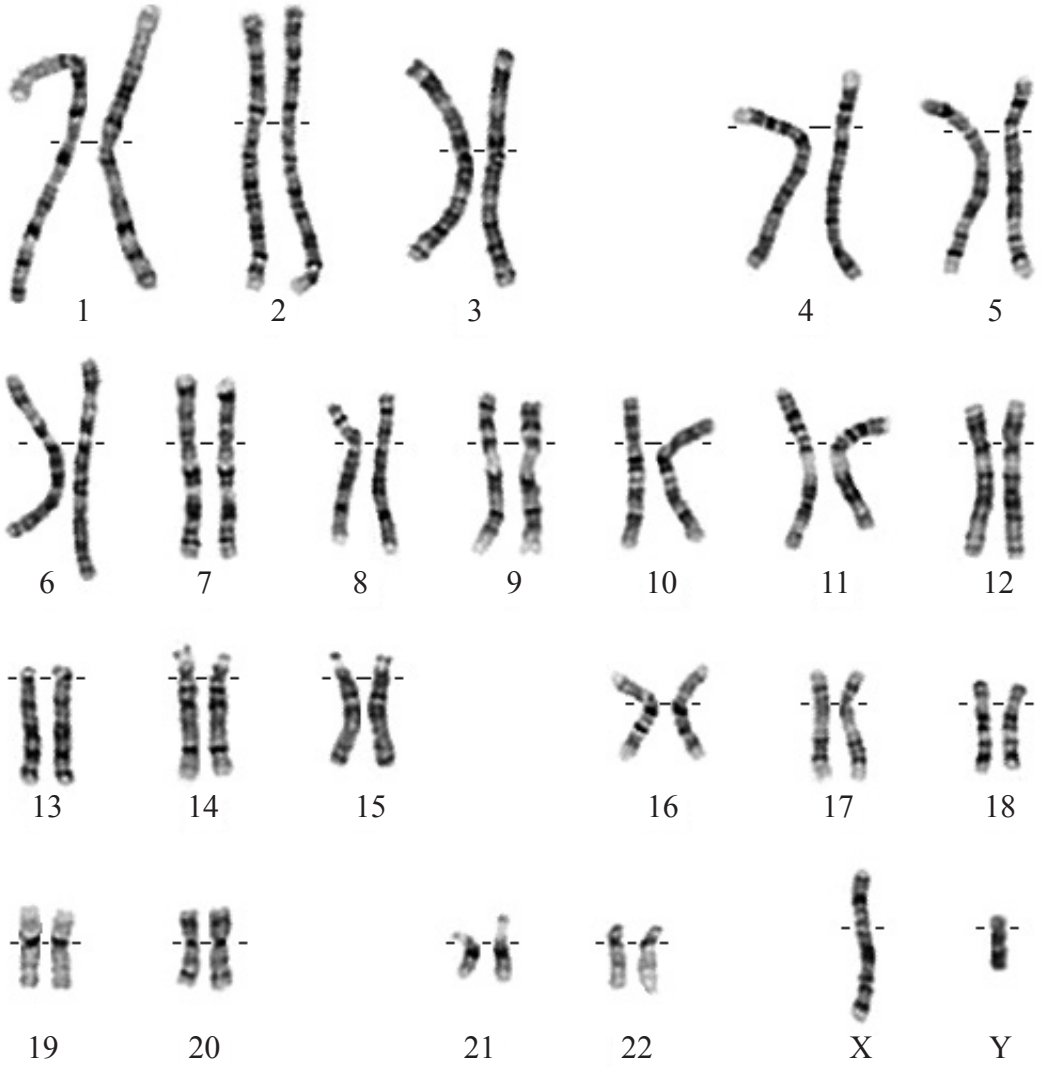
A. How long ago they shared a common ancestor.

B. Which acquired characteristics have been inherited.

C. How different they will be in the future.

D. Which species is most similar to an ancestral form.

15. What is shown by this karyogram?



[Source: www.ucl.ac.uk]

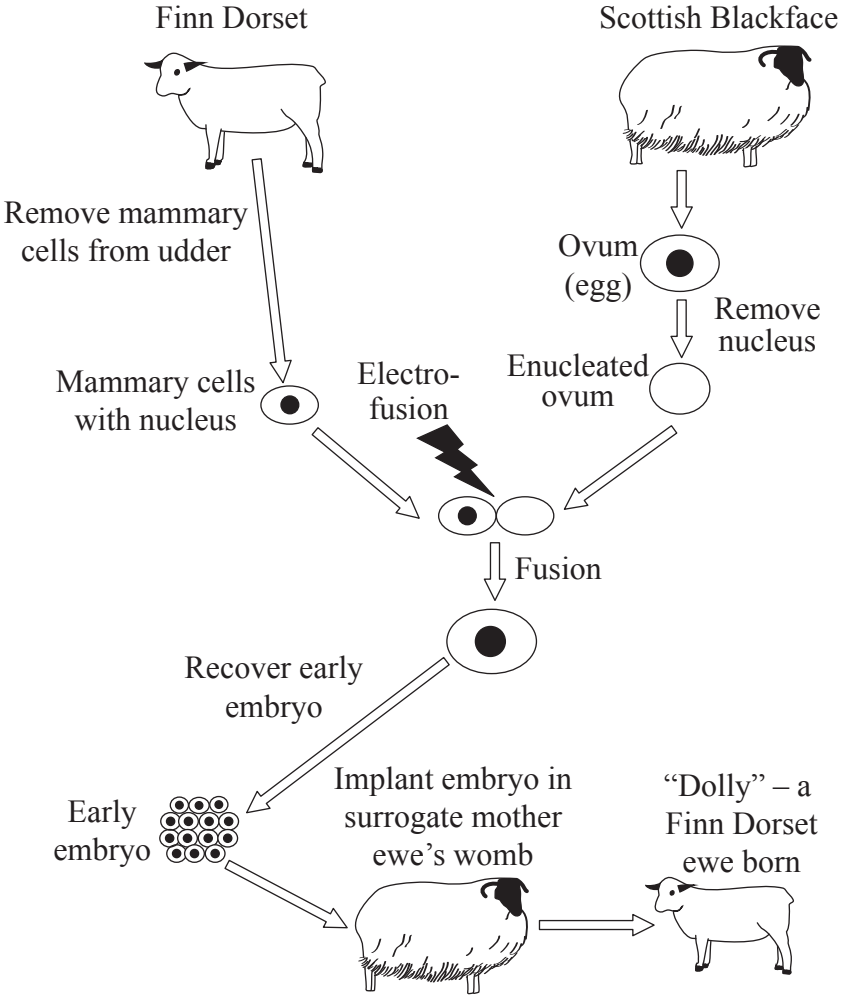
A. The person is male.

B. The person has Down Syndrome.

C. The person has suffered non-disjunction.

D. The person has suffered mutation in chromosome 1.

16. The diagram shows the cloning of the first sheep by Wilmut and Campbell in Scotland in 1997.



[Source: adapted from www.coldmeadow.com]

What is Dolly's DNA composition?

A. The nuclear DNA is from the Finn Dorset and the mitochondrial DNA is from the Scottish Blackface.

B. Half of the DNA is from the Finn Dorset and half is from the Scottish Blackface.

C. All of the DNA is from the Scottish Blackface.

D. All of the DNA is from the Finn Dorset.

17. What is a mesocosm?

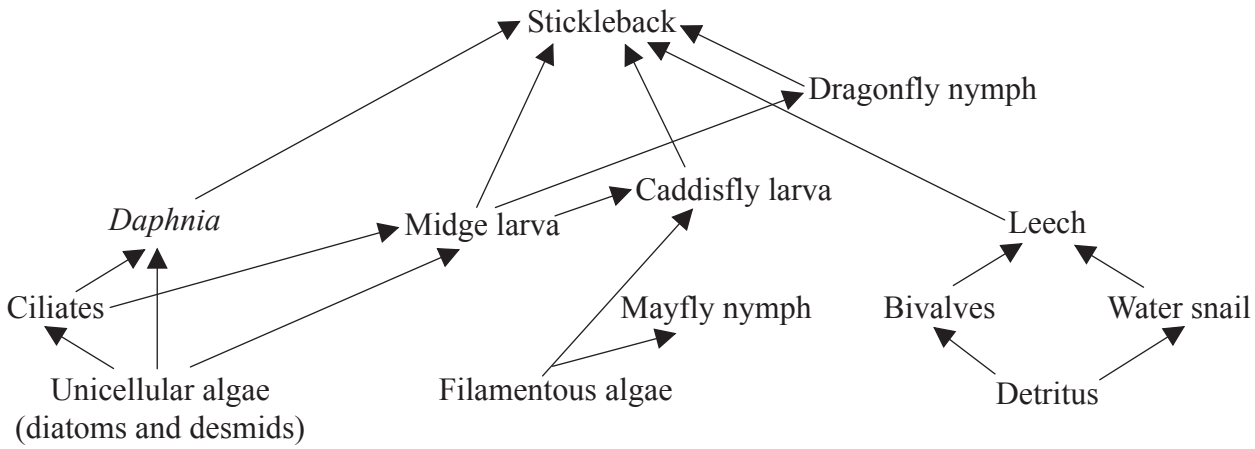
A. A lake where experiments are performed in uncontrolled conditions

B. A small area where parts of the natural environment are kept under controlled conditions

C. An experimental area in a laboratory

D. An ocean

18. The diagram is part of a food web for a freshwater habitat.



Which represents a correct food chain from this web?

A. stickleback → midge larva → unicellular algae

B. ciliates → *Daphnia* → stickleback → dragonfly nymph

C. diatom → midge larva → caddisfly larva → stickleback

D. detritus → bivalves → water snail → leech

19. What is recycled in an ecosystem?

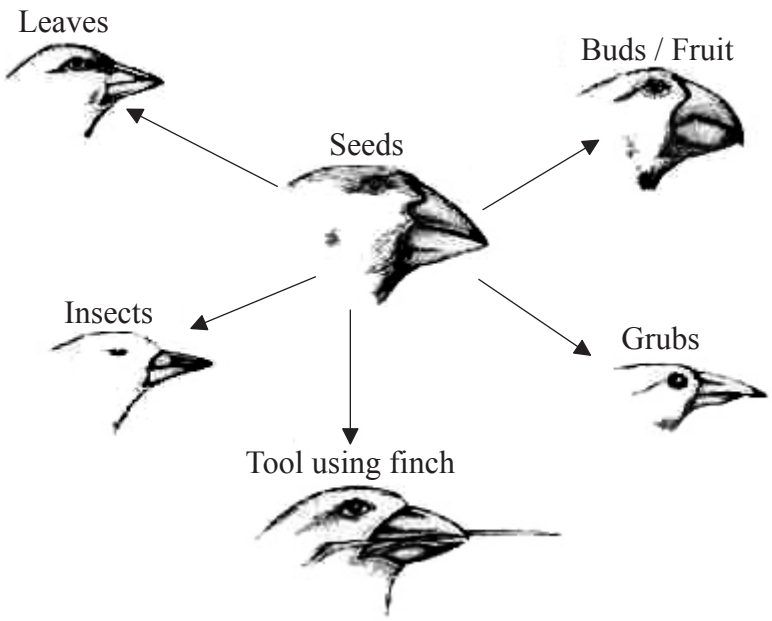
A. Nitrogen, carbon and energy are all recycled.

B. Nitrogen and carbon are all recycled but not energy.

C. Nitrogen is recycled but not carbon or energy.

D. Nitrogen, carbon and energy are not recycled.

20. The diagram shows the beaks of finches on the Galapagos Islands.



[Source: adapted from www.hras.org]

Which process of evolution has occurred in the finches?

A. Convergent evolution of the beaks

B. Natural selection for the strongest beaks

C. Selection pressure caused by the exploitation of different food sources

D. Mutation of the beaks to adapt in different environments

21. What are the three domains of living organisms?

A. Classes, orders and families

B. Bacteria, eukaryotes and viruses

C. Archaea, eubacteria and eukaryotes

D. Decomposers, producers and consumers

22. What characteristic is only found in mammals and **not** other chordates?

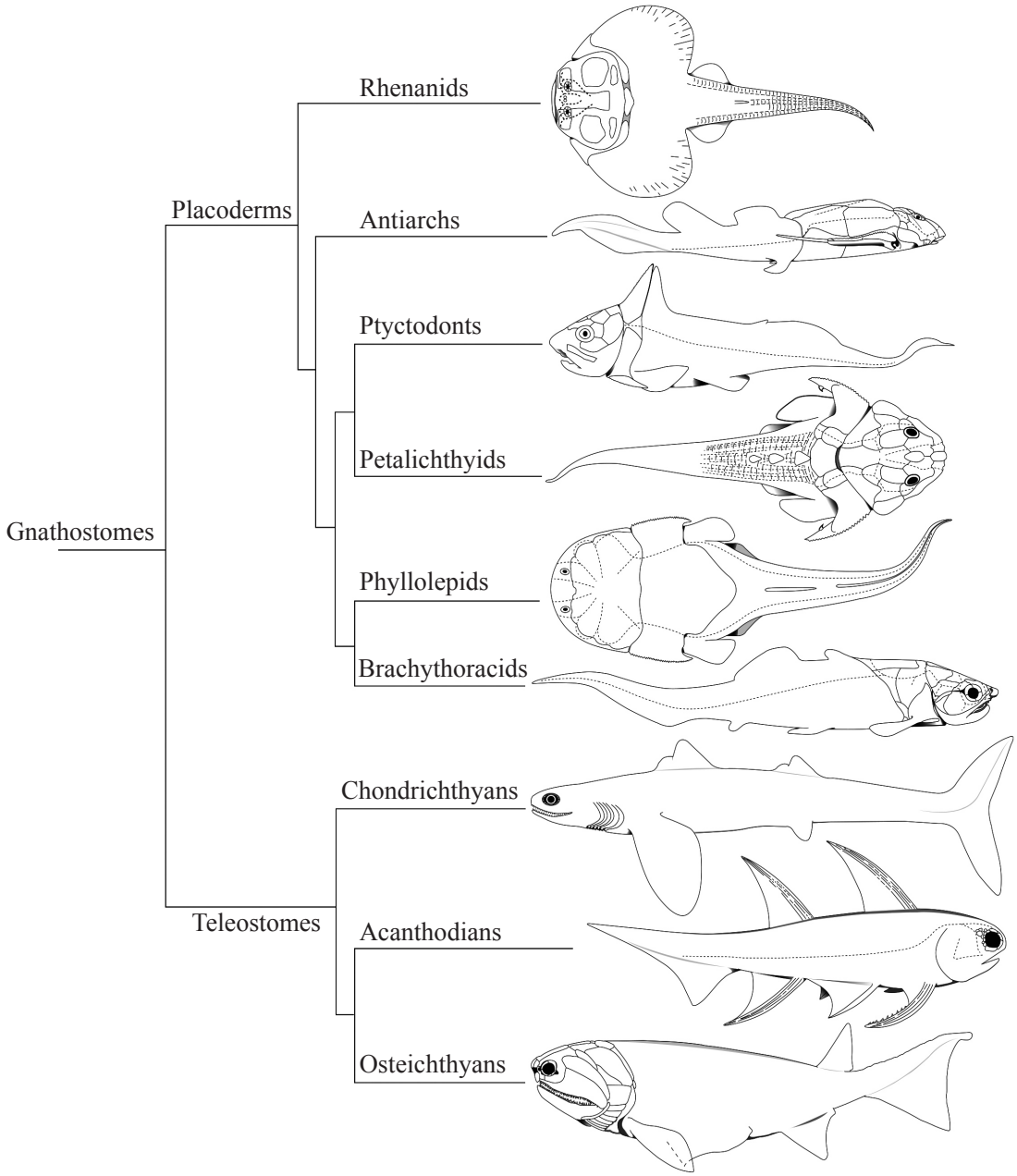
A. Regulation of body temperature

B. Teeth

C. Pentadactyl limb

D. Hair

23. The cladogram shows the phylogenic relationships in jawed vertebrates.



[Source: adapted from R K Carr and G L Jackson, (2008), *Guide to the Geology and Paleontology of the Cleveland Member of the Ohio Shale*, Ohio Geological Survey Guidebook 22, Chapter 5]

What can be deduced about Brachythoracids from this cladogram?

A. They evolved from Placoderms.

B. They gave rise to Gnathostomes.

C. They evolved at the same time as Ptyctodonts.

D. They differ from Phylollepids by only one mutation.

24. What is a characteristic of intestinal villi?

A. They contain few capillaries.

B. They increase the surface area of the small intestine.

C. They have a smooth surface.

D. They are projections from the plasma membranes of the intestinal cells.

25. What is a role of the coronary arteries?

A. To transport blood from the lungs to the heart.

B. To remove deoxygenated blood from the heart muscle.

C. To supply the heart muscle with glucose.

D. To remove cholesterol from the heart muscle.

26. Which statement is a feature of antibodies?

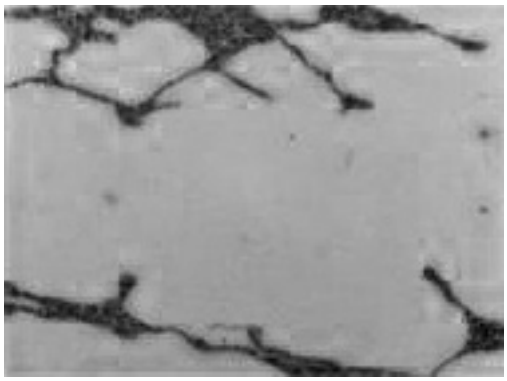
A. Antibodies are pathogenic foreign substances.

B. Antibodies are produced by the bone marrow.

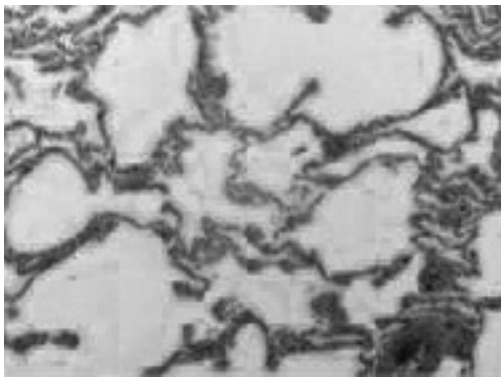
C. Antibodies are composed of polypeptides.

D. Antibodies kill bacteria but not viruses.

27. Emphysema is a long-term, progressive disease that causes shortness of breath. The electron micrographs show the alveoli of a patient with emphysema and the alveoli of a normal person.



Emphysema (magnification × 200)



Normal lung (magnification × 200)

[Source: adapted from www.health-pic.com]

What can be observed in the alveoli of a person suffering from emphysema?

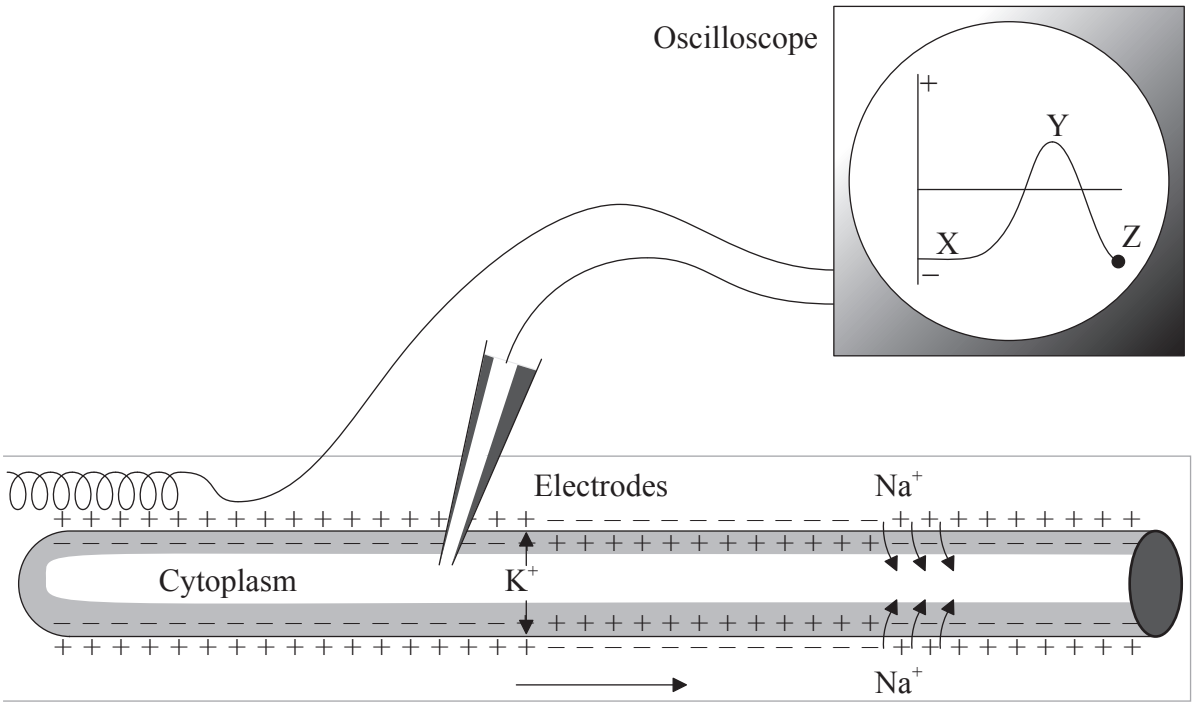
A. Large surface area, large air spaces and many capillaries

B. Small surface area, large air spaces and few capillaries

C. Small surface area, small air spaces and few capillaries

D. Large surface area, small air spaces and many capillaries

28. The diagram shows the results obtained with an oscilloscope attached to a neuron.



[Source: adapted from www.topbiomedical.com/2007/08/action-potential.html]

Why is the change in the oscilloscope occurring between X and Y?

A. Hyperpolarization

B. Hypopolarization

C. Repolarization

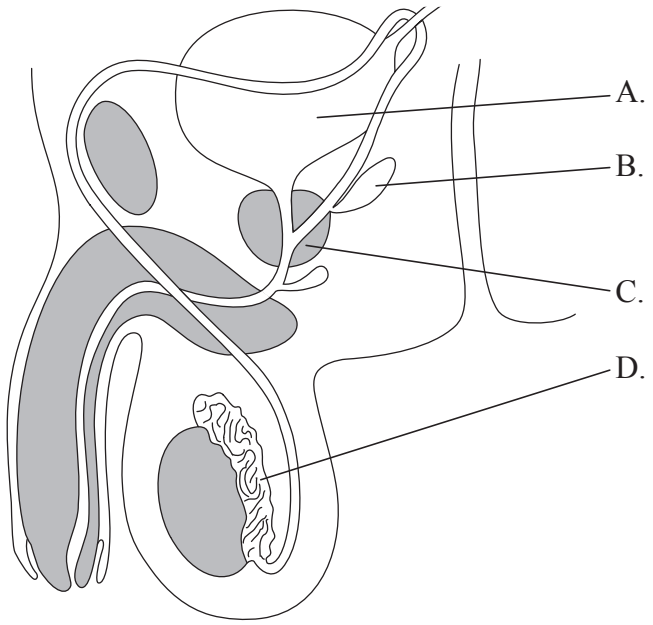
D. Depolarization

29. Which hormone triggers ovulation?

A. FSH

B. Testosterone

C. Progesterone

D. LH

30. The diagram shows a section through the male reproductive system. Where are sperm stored?